

School Fair Prize Draw

Challenge 1

Collect three names and draw a winner.

Build it

1. Create an empty `list` called `entrants`.
2. Use a `for` loop to ask for three names. Use `strip()` and `title()` on each name, then use `append()` to add it to `entrants`.
3. Use `random.choice()` to pick a winner and display "Winner:" followed by the name.
4. Use `random.randint()` to choose 1 to 10 bonus tokens and display "Bonus tokens:" followed by the number.

Test it

- " ana", "BA0" and "cleo " are stored as Ana, Bao and Cleo.
- The winner is one of those names and the tokens are from 1 to 10.

Talk about it

Why must `entrants` be created before the loop?

Save as `challenge1.py`.

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Challenge 2

Let the organiser change the names before the draw.

Build it

1. Create a `list` called `prizes` with three prizes.
2. After the names are collected, ask for a position from 0 to 2 and a new name. Clean the name with `strip()` and `title()`, then put it at that position.
3. Ask for the name of someone who is leaving. Clean it, then use `remove()` to take it out of `entrants`.
4. Display the winner, then a random prize after "Prize:", then the bonus tokens.

Test it

- Names Ana, Bao and Cleo; replace position 1 with "dara"; remove "ana". The winner is Dara or Cleo.
- Only remove a name that is in the `list`. Checking first comes later in the course.

Talk about it

Why must the leaving name be cleaned before `remove()` is used?

Save as `challenge2.py`.

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Challenge 3

Turn the draw into a menu the organiser can use again and again.

Build it

1. Start with three cleaned names in `entrants` and three `prizes`. Store `True` in `running`.
2. Use a `while` loop to display "1: list, 2: add, 3: draw, 4: exit" and ask for a choice.
3. For "1", use a `for` loop to display each entrant. For "2", ask for a name, clean it and append it. For "3", display a random winner, prize and 1 to 10 bonus tokens.
4. For "4", store `False` in `running`. For anything else, display "Choose a valid option."
5. After the loop, use `len()` to display "Entrants remaining:" followed by the count.

Test it

- 2, " eli ", 1, then 4: Eli is listed and "Entrants remaining: 4" appears.
- "x" then 4: "Choose a valid option." then "Entrants remaining: 3".

Talk about it

What makes the `while` loop stop?

Save as `challenge3.py`.